Restaurant Operations Platform

Full-Stack Development | React, Node.js/Express, Firestore, OpenAI GPT-4o, Toast POS, WhatsApp Business Cloud API

Executive Summary

Designed and delivered a full-stack restaurant operations platform (React + Node/Express, Firestore, PM2/IIS) for a multi-location restaurant group, integrating the Toast POS API and the WhatsApp Business Cloud API to automate kitchen, purchasing, catering, and financial workflows. Built AI-powered document intelligence using OpenAI GPT-4o vision to scan supplier receipts into structured, queryable transactions, and real-time, event-driven displays for live order tracking and kitchen operations. Owned the solution end to end — architecture, implementation, deployment, and production support across two restaurant locations.

Key Accomplishments

- **AI receipt-scanning pipeline.** Built an OpenAI GPT-4o vision pipeline that parses uploaded supplier receipts into structured transactions (vendor, totals, line items, unit prices), stored in Firestore with images in Firebase Storage — eliminating manual data entry for the purchasing team.
- **Receipt analytics module.** Developed spend analysis by month, vendor, and item, including item-level price-fluctuation tracking over time with interactive charts, giving owners visibility into cost trends across suppliers.
- **Real-time operational displays.** Engineered event-driven displays (Live Orders, Chef's Kitchen, Order Tracker) backed by Toast POS webhooks and server-sent events, surfacing per-item quantities, wait times, and modifiers to kitchen staff on in-store TVs.
- **WhatsApp automation.** Integrated the WhatsApp Business Cloud API to automate business notifications — customer feedback routing, out-of-stock alerts, daily/monthly reports, and catering invoices — with role- and location-specific recipient routing (owner, manager, chef, purchaser).
- **Catering / party-order automation.** Automated catering operations, including a scheduled job that generates branded PDF prep sheets of next-day orders and delivers them to the relevant chef and manager via WhatsApp the evening before.
- **Intelligent notification batching.** Implemented time-windowed pooling of out-of-stock events into a single consolidated alert, cutting notification noise for staff.
- **Location-aware architecture.** Designed a location-aware configuration layer and reusable services (recipient resolution, scheduling, messaging) so a single codebase serves multiple restaurant locations with per-site behavior.
- **Full deployment lifecycle.** Managed React builds on IIS, the Node service via PM2, environment/secrets management, and production troubleshooting on a live customer-facing system.